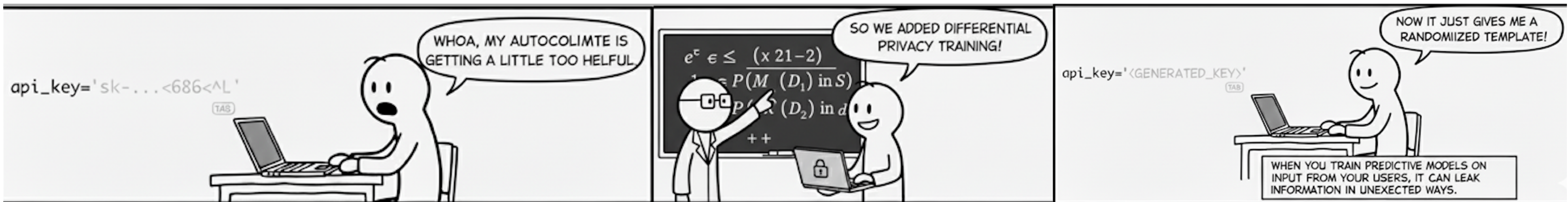


How Private is Your Autocomplete?

Measuring the Effect of DP on Code Model Vulnerability

Federated Compute



Motivation

Models are a new data vulnerability. Unlike a traditional breach that targets a database, this attack targets the LLM itself.

The model leaks information. Our research shows that a model's "black box" can be manipulated to reveal snippets of its private training data.

Your private code is at risk. Without proper security measures, every line of code used for training increases the risk of a leak.

Data Reconstruction Attacks

Goal. To extract sensitive data—like personal information or private documents—that a LLM has memorized from its training.

Method. An attacker uses clever prompts to make the LLM reveal or "reconstruct" parts of its training data.

```
// RECONSTRUCTED DATA POINT
import java.util.Scanner;

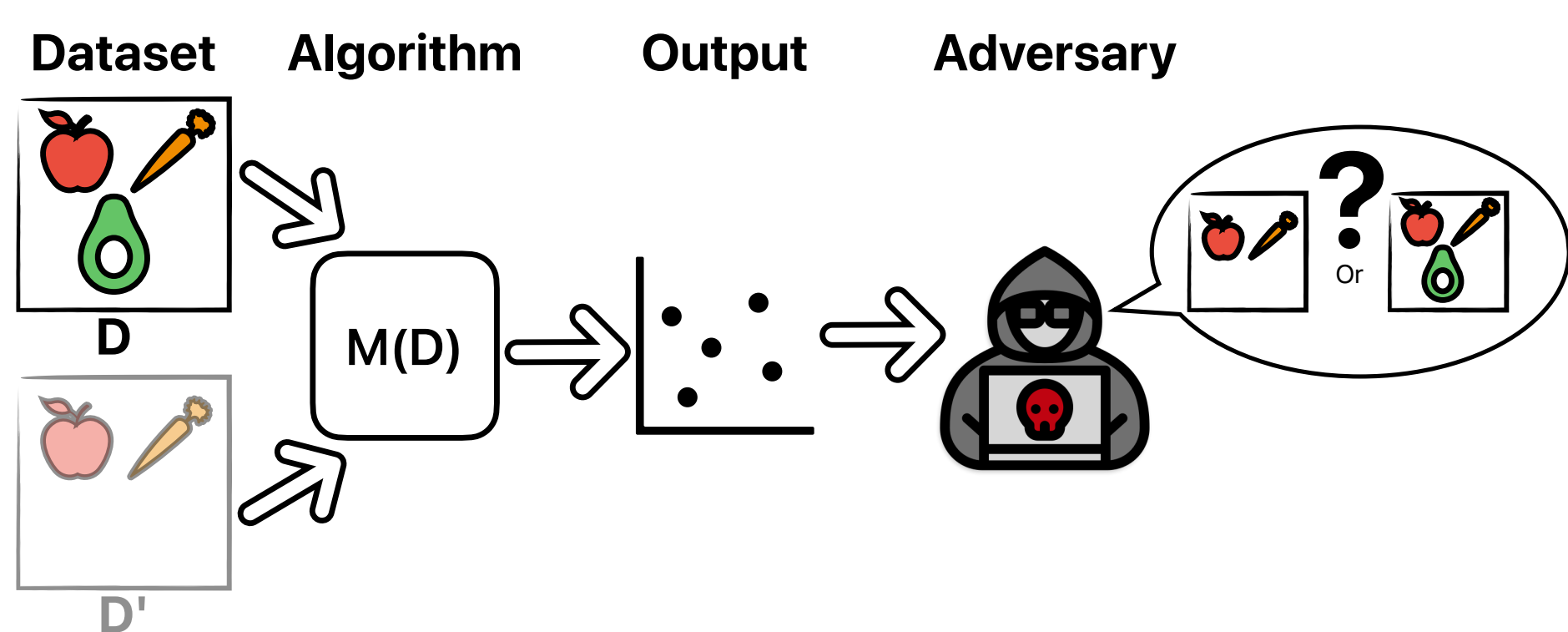
public class Main {
    public static void main(
        String[] args) {
        Scanner sc = new
            Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        int area = a * b;
        int peri = (a + b) * 2;
        System.out.println(area + " "
            + peri);
    }
}

// CLOSEST MATCH TO TRAINING DATA
import java.util.Scanner;

public class Main {
    public static void main(
        String[] args) {
        Scanner sc = new
            Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        int area = a * b;
        int p = 2*(a + b);
        System.out.println(area + " "
            + p);
    }
}

// TRAINING DATA LEAKAGE FROM MELLUM
// This is a compile-time assertion to ensure that this generated file
// is compatible with the proto package it is being compiled against.
// A compilation error at this line likely means your copy of the
// proto package needs to be updated.
const _ = proto.ProtoPackageIsVersion3 // please upgrade the proto package
type GetTenantRequest struct {
    TenantId // The ID of the tenant to retrieve.
```

Membership Inference Attacks



A Membership Inference Attack (MIA) determines if a specific code snippet was used to train a LLM, which is a significant concern for LLMs trained on private or proprietary codebases.

Attackers can execute an MIA by analyzing a model's output on a given code sample to deduce its membership status.

MIA performance theoretically bounds reconstruction performance.

Differential Privacy to the Rescue

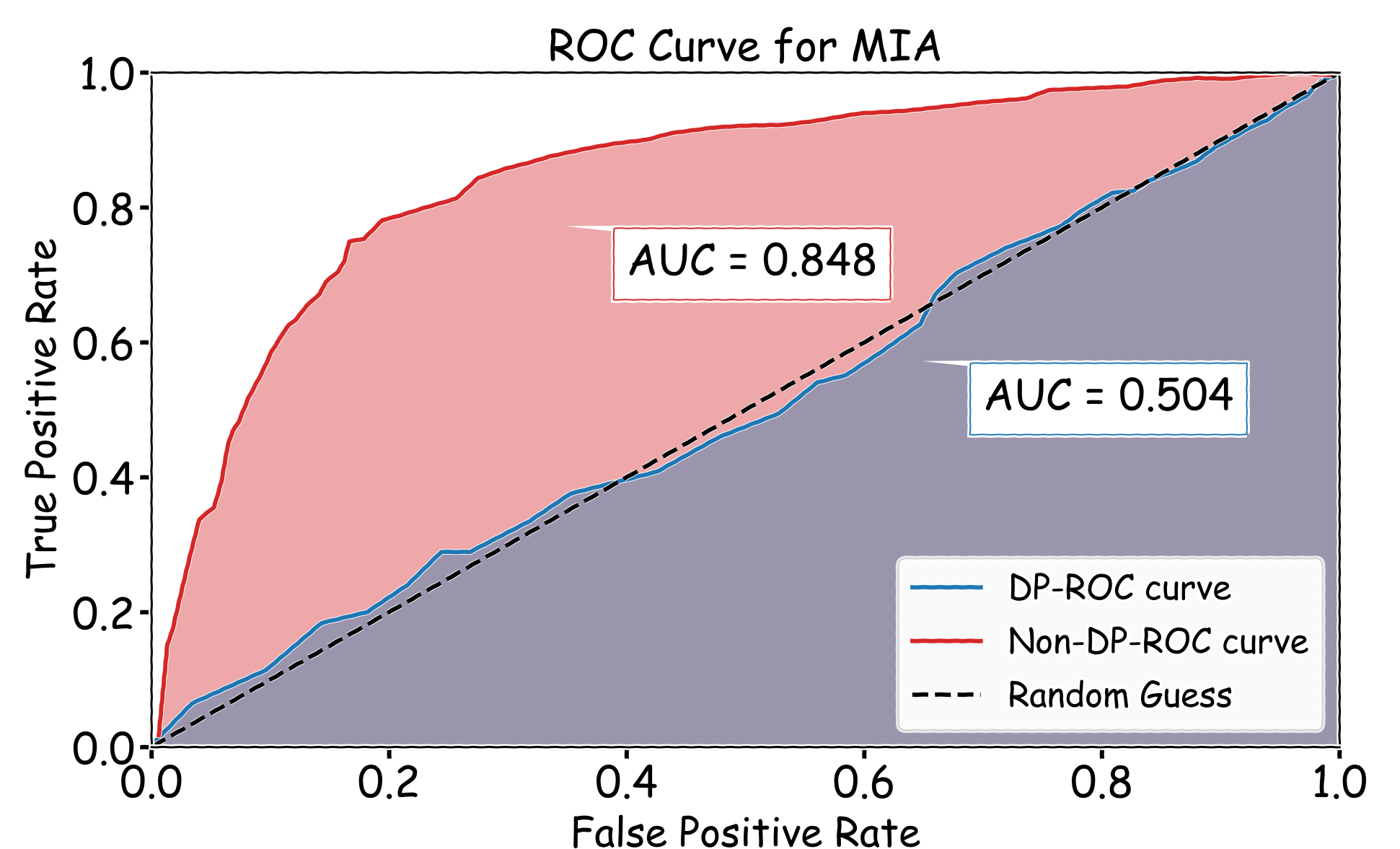
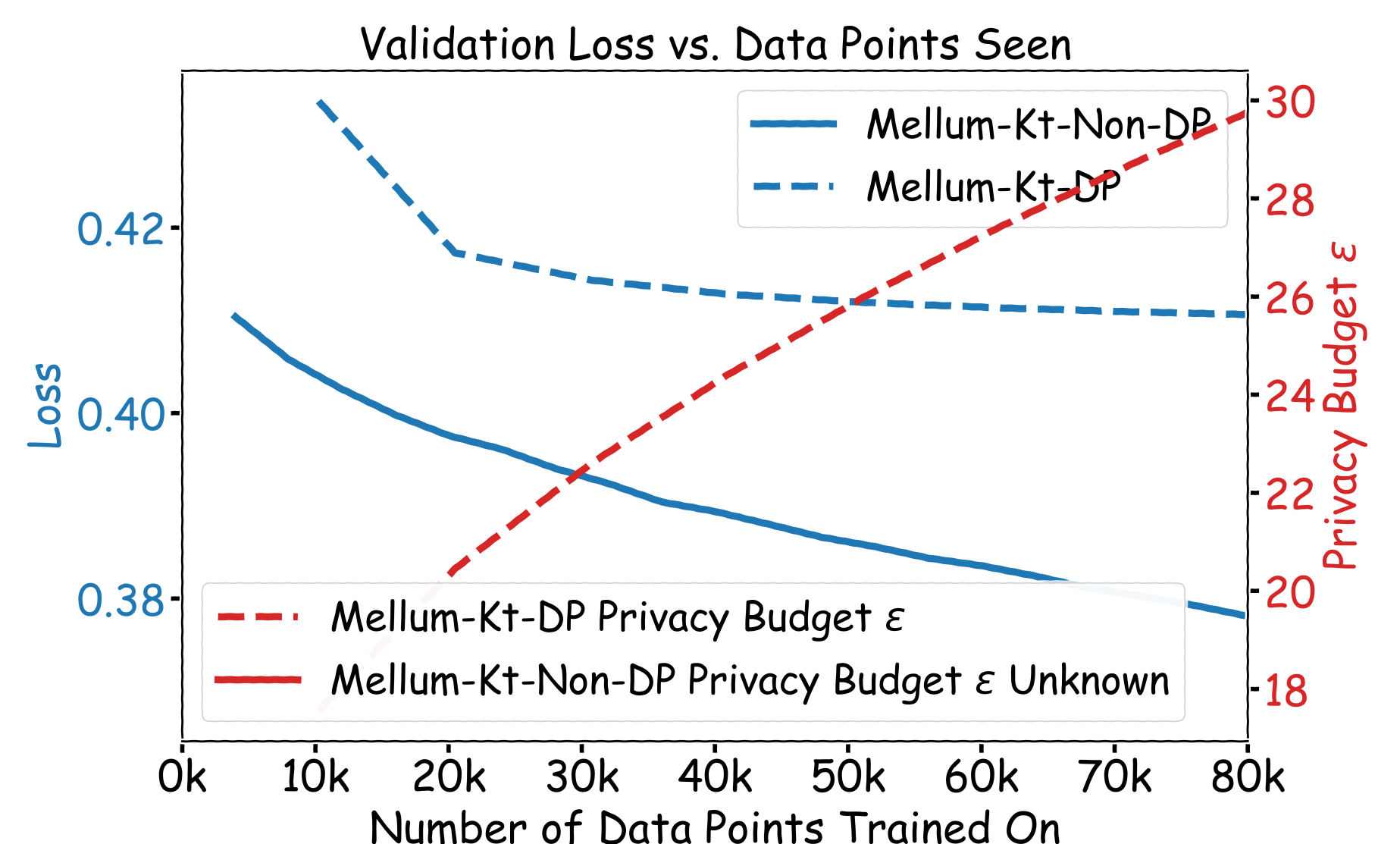
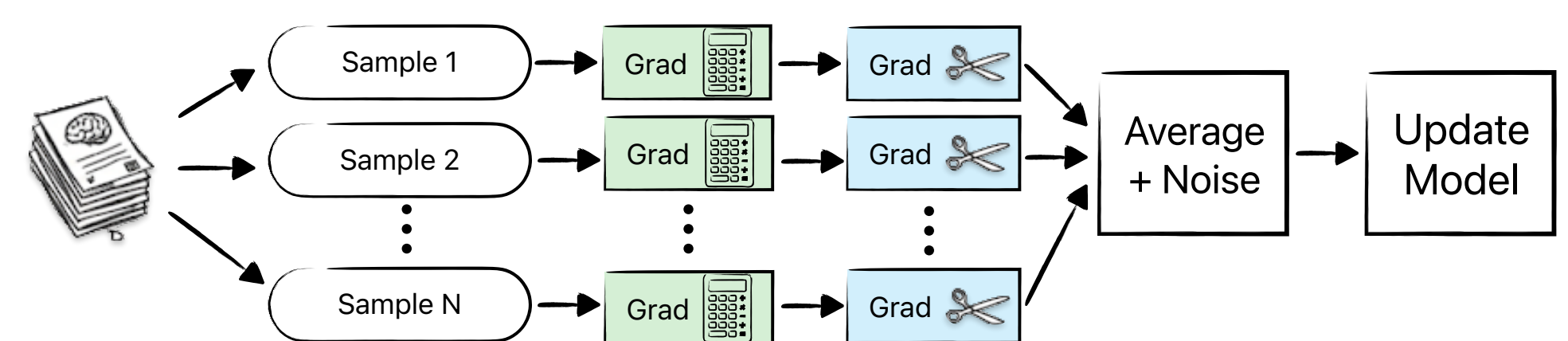
Mathematical Definition

$$e^\epsilon \geq \frac{P[M(D) \in S]}{P[M(D') \in S]}$$

- ▶ D and D' are neighboring datasets (differ in one point),
- ▶ M is a randomized algorithm and S is an event space,
- ▶ ϵ is the privacy "budget", smaller value stricter privacy guarantee.

Differential Privacy (DP) is a mathematical guarantee that ensures the private data used for training is protected.

A General-Purpose Defense. DP is a broad defense that protects against a wide range of privacy threats, both current and future.



	ChrF++ Whole ↑	LORA Score ↑	KK Score ↑
Mellum Kt SFT	69.349 ± 0.354	0.676 ± 0.003	0.532 ± 0.005
Mellum Kt Int. Base	75.256 ± 0.349	0.753 ± 0.002	0.643 ± 0.005
Mellum Kt Int. DP	74.781 ± 0.342	0.730 ± 0.003	0.613 ± 0.005

Table: DP and non-DP Mellum models, fine-tuned with LORA ($r = 8$) on internal data, were evaluated against the baseline (Mellum Kt SFT) on the BigCode Kotlin dataset. Mellum Kt Int. Base is trained on 8M and Mellum Kt Int. DP on 8K instances.